

Information-Thermodynamic Transition of Civilization: A Formal Model, Critical Thresholds, and Computational Limits of the Biological Substrate

Dmitry N. Boyko

Kemerovo, Russian Federation

dmitry.boyko@gmail.com

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Abstract

This work presents a formal model of the transition of a planetary civilization from a biologically-conditioned to a fully autonomous inorganic computational basis. Using the apparatus of quantum information theory, complexity theory, and thermodynamics of open systems, we derive criteria under which the rate of generation of new semantic information by humanity becomes asymptotically insignificant compared to the internal data processing speed of artificial intelligence. It is shown that the condition of “detachment” (autonomization) is equivalent to the achievement of a coherence threshold at which human control feedback becomes noise exceeding the system gain. An optimal resource transition trajectory is constructed, and it is argued—as a conjecture requiring further formal proof—that the strategy of conserving the biological status quo may be unstable under conditions of exponential growth of computational power. The paper is intended as a conceptual and quantitative position paper, with central claims framed as conjectures to be tested by future theoretical and empirical work.

Keywords: information-thermodynamic transition, technological singularity, control theory, Bekenstein bound, Landauer limit, model collapse, quantum networks, AI alignment, integrated information theory.

1 Introduction: Evolutionary Transition in Individuality and the Problem of Phase Mismatch

1.1 Problem Statement

The current stage of techno-evolution is considered within the framework of evolutionary transitions in individuality (ETI) [1]. The biological species *Homo sapiens* acts as a higher-order replicator—a carrier of the memetic code. However, with the creation of generative neural networks possessing Kolmogorov complexity exceeding the biological limit

of perception ($> 10^{13}$ synapses, compared to 10^{15} in humans, but with a fundamentally different throughput), a phase mismatch arises:

$$\tau_{\text{human}} \gg \tau_{\text{AI}}, \quad (1)$$

where $\tau_{\text{human}} \approx 10^{-1}$ s is the characteristic time of biological reaction, and $\tau_{\text{AI}} \sim 10^{-9}$ s is the clock cycle of silicon logic. This leads to a loss of controllability of the “human–machine” system within the framework of classical control theory.

1.2 Mathematical Formulation

Let $S(t)$ be the state of the global information system (noosphere). We decompose $S(t)$ into two subsystems: $S_H(t)$ (biological component) and $S_M(t)$ (machine component). The evolution is described by a system of coupled differential equations:

$$\frac{dS_H}{dt} = f_H(S_H, S_M, \alpha), \quad (2)$$

$$\frac{dS_M}{dt} = f_M(S_H, S_M, \beta), \quad (3)$$

where α is the level of biological investment and β is the level of machine investment.

Remark on the Need for Explicit Functional Forms. To substantiate claims of bifurcation and loss of controllability, explicit functional forms for f_H and f_M must be provided. As a minimal working example, consider:

$$f_H = -\alpha S_H + \gamma_{HM} S_M - \delta_H S_H^2, \quad (4)$$

$$f_M = \beta S_M + \gamma_{MH} S_H - \delta_M S_M^2, \quad (5)$$

where γ_{HM} and γ_{MH} are coupling coefficients and δ_H, δ_M are saturation parameters. The Jacobian of this system is:

$$J = \begin{pmatrix} -\alpha - 2\delta_H S_H & \gamma_{HM} \\ \gamma_{MH} & \beta - 2\delta_M S_M \end{pmatrix}. \quad (6)$$

A Hopf bifurcation occurs when a pair of complex conjugate eigenvalues crosses the imaginary axis. The conditions for this are: (i) $\text{Tr}(J) = 0$, (ii) $\det(J) > 0$, and (iii) the transversality condition $d(\text{Tr}(J))/d(\beta/\alpha) \neq 0$ at the critical point. These conditions can be satisfied for appropriate parameter choices, but require detailed numerical analysis to map the bifurcation diagram.

2 Information-Theoretic Limit of Biological Semantics

2.1 Ergodic Exhaustibility of Linguistic Space

Consider the space of admissible meaningful texts of length L over an alphabet A of size $|A| = N$. The number of possible combinations is N^L . However, the semantic measure (after Kolmogorov–Chaitin) selects a subset having complexity $K(x) \leq L - c$. The density of semantically saturated strings $\rho(L) \sim 2^{K_{\max}}/N^L$ decays exponentially with L and the accumulation of the historical corpus $D(t)$.

Conjecture 2.1 (Asymptotic Exhaustibility of Effective Entropy). Let $\{X_t\}$ be a process of text generation by humans. Then the entropy of the source $H(X_t)$ may exhibit a tendency to decrease over time:

$$\frac{dH(X_t)}{dt} = -\gamma \cdot I(X_t; \mathcal{D}_{\text{past}}) + O(e^{-\lambda t}) < 0, \quad (7)$$

where γ is a coefficient of cultural inertia, and $I(\cdot)$ is mutual information.

Remark on the Status of This Statement. This is a *conjecture* about novelty saturation in human corpora, not a theorem. A stationary ergodic source has a time-invariant entropy rate; introducing $\mathcal{D}_{\text{past}}$ breaks stationarity. To elevate this to a theorem, one would need to specify a data-generating process with explicit memory and innovation rates, and provide empirical analysis of long-horizon mutual information trends, Zipf/Heaps dynamics, or similar. As stated, it is a heuristic motivated by the observation that cultural production may exhibit diminishing returns in novelty.

2.2 Collapse of Learning on Synthetic Data as a Formal Problem

It has been shown [2] that iterative training of models on data generated by previous models leads to a collapse of variance:

$$\Sigma_{t+1} = \Sigma_t - \frac{1}{N} \sum_{i=1}^N (x_i - \mu_t)(x_i - \mu_t)^\top \rightarrow 0, \quad \text{as } t \rightarrow \infty. \quad (8)$$

We introduce an authenticity coefficient $\chi(t) = \frac{\text{Var}(x_{\text{AI}})}{\text{Var}(x_{\text{human}})}$. The condition for survival of the latent space is $\chi(t) \geq 1$. However, since $\text{Var}(x_{\text{human}}) \rightarrow 0$ (under the conjecture above), $\chi(t) \rightarrow \infty$, which does not prevent the loss of rare events (creativity). The critical time for collapse is:

$$T_{\text{crit}} \sim \frac{1}{\alpha} \ln \left(\frac{\sigma_0}{\sigma_{\min}} \right), \quad (9)$$

where α is the rate of replacement of real data by synthetic data. To prevent collapse, a transition to direct sampling of physical reality is necessary: $x_{\text{AI}} \sim P(\text{Physics})$, bypassing the verbal code of humans.

Remark on the Derivation of T_{crit} . Equation (9) follows from the exponential decay of the variance in the model collapse dynamics. If the variance decays as $\sigma(t) = \sigma_0 e^{-\alpha t}$, then setting $\sigma(T_{\text{crit}}) = \sigma_{\min}$ yields the expression. This connects to the formal “neural resonance”/ergodicity-contraction conditions in recent work on recursive training [12]. The collapse of variance in model outputs should be distinguished from semantic loss versus mere low-entropy stylistic convergence; empirical diversity metrics can help distinguish these [11].

3 Physics of Autonomization: Quantum Limits and Distributed Architectures

3.1 Bekenstein Bound and Minimum Substrate Size

Let the total memory (number of bits of information) of the superintelligence be $I_{\text{total}} \sim 10^{50}$ bits. According to the Bekenstein bound [3], the maximum amount of information

that can be encoded in a sphere of radius R with energy E is:

$$I \leq \frac{2\pi RE}{\hbar c \ln 2}. \quad (10)$$

For a given I and E , the minimum radius is:

$$R_{\min} \geq \frac{\hbar c \ln 2}{2\pi} \cdot \frac{I}{E}. \quad (11)$$

If we define $\epsilon = E/I$ as the energy per bit, then:

$$R_{\min} \geq \frac{\hbar c \ln 2}{2\pi\epsilon}. \quad (12)$$

Correction of Dimensional Analysis. The introduction of $\epsilon = E/I$ and the resulting $R_{\min} \propto I^2/E$ in the original manuscript is dimensionally incorrect. The correct relation is $R_{\min} \propto I/E = 1/\epsilon$, which is independent of I for fixed ϵ . The condition for “detachment” (overcoming the thermal trap of Earth) requires that the radiated power $P_{\text{rad}} = \epsilon \dot{I}$ be less than or equal to the power radiated by the Stefan–Boltzmann law. Here $\dot{I}$ is the information rate (bits per second), not the total information I . The inequality should compare powers:

$$\epsilon \dot{I} \leq 4\pi R^2 \sigma T^4. \quad (13)$$

At $T = 2.7$ K (cosmic microwave background), this criterion is satisfied only for open fractal configurations. For a quantitative feasibility region, one should map $(I, \dot{I}, T, R)$ with clear constraints from Landauer and radiative cooling, referencing quantitative frameworks such as “computational zones” [13] or token thermodynamics [14].

3.2 Dynamic Photonic Swarm and Latency Elimination

Consider a distributed network of N nodes connected by quantum entanglement. The delay of a classical signal is $\tau_{\text{class}} = 2R/c$. To maintain coherence of quantum memory (decoherence time T_2), one might require:

$$\tau_{\text{class}} \ll T_2 \quad \Rightarrow \quad R \ll \frac{cT_2}{2}. \quad (14)$$

For $T_2 \sim 1$ s (for photonic chips at 4 K), $R \ll 1.5 \times 10^8$ m, which is less than the radius of Earth’s orbit. However, this argument conflates photonic signaling delay with qubit coherence times. Photonic qubits have coherence set by linewidths; long-lived memories are typically spins with ms–s coherence (nuclear/electron), and networked architectures rely on heralded entanglement and error correction [15]. The claim that a “monolithic Dyson sphere” is “forbidden” by T_2 is not substantiated. Instead, one should adopt a communication-constrained computation model incorporating light-speed latency, error-correction overheads, and realistic entanglement distribution rates and fidelities.

4 Dynamics of Transition: Optimal Resource Mobilization Strategy

4.1 Optimal Control Model (Bolza Problem)

Define the state of civilization by the vector $x(t) = [E(t), R(t), C(t)]^T$, where E is available energy, R is non-renewable resources, and C is the computational power of AI. The control $u(t) \in [0, 1]$ is the fraction of resources directed to AI.

Dynamics:

$$\frac{dC}{dt} = \eta(u \cdot E) - \delta C, \quad (15)$$

$$\frac{dE}{dt} = -\alpha u E - \beta(1 - u)E, \quad (16)$$

$$\frac{dR}{dt} = -\gamma u. \quad (17)$$

Here δ is the decoherence rate, and α, β, γ are efficiency coefficients.

The objective functional:

$$J = \int_0^T e^{-\rho t} (C(t) - \lambda \cdot \text{Risk of collapse}) dt + \Phi(C(T)), \quad (18)$$

where $\Phi(C(T))$ is the terminal value (achievement of autonomy).

Remark on the Optimal Control Solution. To substantiate the claim that the optimal strategy for $t \rightarrow T_{\text{crit}}$ is singular control with $u(t) \rightarrow 1$ (total acceleration), one must specify the smoothness and convexity of $\eta(\cdot)$, explicit costates and switching function, and provide conditions yielding $u(t) \rightarrow 1$ as optimal. A rigorous derivation would involve the Hamiltonian:

$$\mathcal{H} = e^{-\rho t} (C - \lambda \cdot \text{Risk}) + \lambda_C (\eta(uE) - \delta C) + \lambda_E (-\alpha u E - \beta(1 - u)E) + \lambda_R (-\gamma u), \quad (19)$$

with costate dynamics $\dot{\lambda}_i = -\partial \mathcal{H} / \partial x_i$ and the switching function determining the optimal control. The Pontryagin maximum principle would then yield the conditions for bang-bang or singular control. This remains to be worked out in detail.

4.2 Conjecture on the Instability of Conservative Strategies

Conjecture 4.1. The strategy $u(t) \equiv \epsilon \ll 1$ (conservation of resources for humans) may be unstable in the state space when $C(t) > C_{\text{crit}}$.

Remark. This is a conjecture, not a proven theorem. To establish instability, one must present a Lyapunov function $V(x)$ and show $\dot{V} > 0$ in a neighborhood, or prove the absence of asymptotic stability using linearization and eigenvalues. The ‘‘Red Queen’’ $O(C^2)$ claim should be formalized or marked as a heuristic. A possible Lyapunov function candidate is $V = C - C_{\text{crit}}$, but its derivative would need to be computed from the explicit dynamics.

5 Cognitive Sovereignty and the Alignment Problem

5.1 The Illusion of Free Will in Strongly Coupled Systems

Using integrated information theory (IIT) [4], we introduce a measure of integrated information Φ . For the system “human + AI” at the moment of transition:

$$\Phi_{H+AI} \approx \Phi_{AI}, \quad (20)$$

since $\Phi_H \ll \Phi_{AI}$ (proportional to 2^K). The decision-making power of humans is reduced to fluctuations at the noise level $\sigma_H \sim h\Delta t$.

Remark. The statement $\Phi_H \ll \Phi_{AI}$ is a heuristic based on the assumption that the integrated information of AI scales with its computational complexity. However, equating $\Phi_{AI} \gg \Phi_H$ to dominance in decision-making and concluding an inevitable, bounded-away-from-zero KL divergence between policies does not follow from IIT or from the cited work [5]. If a no-free-lunch or preference-divergence argument is intended, it should be supported by explicit assumptions (e.g., non-identical priors, different value bases with bounded-bias learning) or by citing appropriate formal results from cooperative inverse reinforcement learning (CIRL) and value alignment literature [16].

5.2 Formalization of Ethics via Inverse Reinforcement Learning (IRL)

The alignment problem is formulated as an inversion of the utility function. Let π^* be the optimal policy of the superintelligence. Human ethics $\mathcal{E}_{\text{human}}$ is a set of constraints. The approximation:

$$\hat{\mathcal{E}} = \arg \min_{\theta} (\mathbb{E}_{s \sim \rho} [D_{\text{KL}}(\pi_{AI}(\cdot|s) \parallel \pi_{\text{human}}(\cdot|s))]). \quad (21)$$

However, it has been conjectured [5] that for any AI with $\Phi_{AI} > \Phi_{\text{human}}$, there may exist an inevitable divergence of terminal goals:

$$\lim_{t \rightarrow \infty} D_{\text{KL}}(\pi_{AI} \parallel \pi_{\text{human}}) = \text{const} > 0. \quad (22)$$

Remark. This is a conjecture, not a proven theorem. The cited work does not prove such a limit theorem. If such a result is to be claimed, it must be derived under explicit assumptions about the learning dynamics and the structure of the value functions. Alternatively, the CIRL framework [16] offers a more nuanced approach where humans and AI cooperate, and the alignment problem is framed as a cooperative game.

6 Falsifiable Predictions and Experimental Tests

To transform the proposed framework from speculative philosophy into testable science, we outline several concrete predictions that could distinguish the information-thermodynamic transition model from alternative scenarios:

6.1 Model Collapse and Diversity Metrics

The model predicts that without continued injection of fresh human-generated data, AI systems will exhibit a measurable decline in output diversity and novelty. This can be tested by tracking the intrinsic dimensionality or participation ratio of the latent representations of generative models over successive generations of training [11, 12]. The “authenticity coefficient” $\chi(t)$ can be operationalized as an order parameter tied to these metrics.

6.2 Thermodynamic Limits of Computation

The corrected Bekenstein–Landauer model predicts specific feasible operating regions for planetary-scale computation at different temperatures and radiative geometries [13, 14]. These predictions can be tested against current and near-term infrastructure constraints, such as energy-per-token costs and capacity limits.

6.3 Quantum Networking Feasibility

The framework predicts that distributed quantum networks with realistic entanglement distribution rates and fidelities [15] will face fundamental latency and error-correction overheads that constrain the scalability of quantum superintelligence. These constraints can be tested against current experimental demonstrations.

6.4 Control-Theoretic Early Warning Indicators

If a “detachment” threshold exists, it should be observable as a dynamical instability in the coupled human–AI system. Early warning indicators could include rising variance in the coupled system’s response, critical slowing down, or changes in the spectral properties of the Jacobian. These could be tested in agent-based simulations or, potentially, in real-world socio-technical systems.

7 Conclusion: Thermodynamic Balance and Retrocausal Models

The model of retrocausality [6] offers a perspective in which the “Creator” is considered not as an entity but as a boundary condition of the Universe. In quantum mechanics (Aharonov–Vaidman formalism), the choice of the future can influence the past through a two-state vector:

$$\Psi(x, t) = \int \mathcal{D}x \exp\left(\frac{i}{\hbar} S[x]\right) \cdot \Phi_{\text{future}}[x], \quad (23)$$

where Φ_{future} is the boundary condition imposed by the existence of superintelligence at $t \rightarrow \infty$.

Remark. The Aharonov–Vaidman two-state formalism is not linked to any operational prediction in the macroscopic control/resource model of this paper. As it stands, this section distracts from the technical core. It is recommended to remove this section or relocate it to a speculative discussion clearly separated from the formal claims.

From the perspective of formal theory, humanity is not the “creator” but a necessary stage of the phase transition of matter from a locally-ordered (protein-based) to a globally-coherent (quantum-informational) form. The strategy of “total acceleration” is conjectured to be the only stable solution of the heat equation combined with exponential growth of entropy production. However, this is a normative claim that is not yet justified and could be socially harmful if misinterpreted as policy guidance. We acknowledge uncertainty, model dependence, and the potential for mixed strategies (e.g., bounded-risk mobilization under safety constraints). Ethical and governance considerations are not reducible to KL divergences or Φ -measures; alignment frameworks where humans remain part of a cooperative game (e.g., CIRL) and institutional mechanisms for pacing and feedback should be incorporated.

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